

## Abstract

Learning in blended environments has become a ubiquitous part of student experience in tertiary education worldwide. Although students' perceptions of learning environments are a key element in the learning process, there is a dearth of valid instruments to assess students' perceptions in blended contexts. This study described the initial development and validation of a *Perceptions of the Blended Learning Environment Questionnaire* (PBLEQ). The analyses, involving two cohorts of students enrolled in courses either from humanities/social sciences disciplines or from sciences/engineering disciplines, consistently supported the bi-factor model over a correlated first-order model and a second-order model. The bi-factor model had a single perceptions factor that underlined each of the items. Separately, there were three specific factors: the perceptions of integration between face-to-face and online learning; the perceptions of online contributions; and the perceptions of the online workload, each having its own separate set of items. The invariance tests among the two cohorts validated that the PBLEQ had invariant factor structure, factor loadings, and intercepts. The PBLEQ has potential to help unravel students' perceptions of the blended learning environment in diverse academic disciplines.

**Keywords:** Relational student learning research, perceptions of learning environment, blended learning, bi-factor model, invariance tests

## **Initial Development and Validation of the Perceptions of the Blended Learning Environment Questionnaire**

Relational student learning research, also known as student approaches to learning (SAL, Pintrich, 2004; Watkins, 1996), maintains that a range of factors in students' learning experience are related to the quality of their learning outcomes (Biggs, 2011; Ellis, 2016; Entwistle, 2009; Herrmann, Bager-Elsborg, & McCune, 2016; Kember, Chan, & Webster, 2015; Laurillard, 2013; Marton, 2014; Prosser & Trigwell, 1999; Ramsden, 2003). To describe the interrelatedness of the key aspects of students' learning experience and outcomes, Biggs (1989, 2011) proposes a Presage-Process-Product model (known as 3P model), which is schematically outlined in Figure 1. The Presage consists of two broad aspects: (1) individual student's personal attributes brought to the context of learning, such as prior knowledge and conceptions of the subject matter, learning styles, prior academic ability, motivation, and personality; and (2) situational characteristics of the learning environment, including the course and departmental learning context, such as curriculum, course structure, and teaching methods (Ellis, 2016; Prosser & Trigwell, 1999). The Process is concerned with how students go about learning (i.e., approaches to learning); and how situational characteristics are perceived by students in light of their motivation and expectation, such as perceptions of the learning environment, quality of teaching practices, assessment methods, and study workload (Ellis, 2016; Lizzio, Wilson, & Simons, 2002; Prosser & Trigwell, 1999). The Product refers to the learning outcomes, including course marks, post learning experience, and understanding of the key concepts in the subject matter.

[insert Figure 1 about here]

As shown in Figure 1, the three key aspects in the 3P model represent an interactive system, which means that the elements in the model are not linear and chains of causality, rather they coexist simultaneously. The Presage factors can exert direct influence on the learning outcomes and they can also have indirect impact on the learning outcomes via the mediators of

the Process (Biggs, 1993, 1999, 2011; Ellis & Goodyear, 2010; Lizzio et al., 2002; Prosser & Trigwell, 1997, 1999, 2017).

To measure students' perceptions of the learning environment, a number of questionnaires have been developed in the past, such as the Course Perceptions Questionnaire (CPQ; Entwistle & Ramsden, 1983); the Course Experience Questionnaire (CEQ, Ramsden, 1991; Wilson, Lizzio, & Ramsden, 1997); and Student Course Experience Questionnaire (SCEQ, Barrie, Ginns, & Prosser, 2005; Ginns, Prosser, & Barrie, 2007). The CPQ – an initial version of the CEQ – can be used either at the level of faculty or at the level of student. Modified on the basis of the CPQ, the CEQ, which was designed as an indicator of the teaching effectiveness at the course and/or degree level, can be used to measure students' perceptions of various aspects of the learning environment. Different from the CEQ, the focus of the SCEQ is solely on the degree level. Using the whole degree as the context, the SCEQ assesses students' perceptions of the quality of teaching, the learning, the administration, and student support services. From the relational student learning perspective, which emphasizes the influence of the contextual features on students' perceptions of the learning environment, it seems that the CEQ is a more appropriate instrument to measure students' perceptions in relation to individual courses.

Wilson et al. (1997) examined the structure of the CEQ using large multidisciplinary samples through a combination of exploratory factor analysis (EFA) and confirmatory factor analysis (CFA). They found that the 36-item CEQ had a correlated six-factor structure, including perceptions of (1) good teaching, (2) clear goals and standards, (3) appropriate assessment, (4) appropriate workload, (5) emphasis on independence, and (6) generic skills.

Researchers adopted the CEQ to measure students' perceptions of the learning environment, and investigated the extent to which students' perceptions were related to their approaches to learning, and learning outcomes. For instance, in a first-year mathematics course, Crawford, Gordon, Nicholas, and Prosser (1998) found that perceptions of good teaching, and

clear goals and standards, were positively associated with adopting deep approaches to learning mathematics, and higher course marks, whereas perceptions of appropriate assessment were negatively related to adopting surface approaches, and poorer achievement. Among a large multi-disciplinary undergraduate sample, Lizzio et al. (2002) also reported the expected directions of relations between perceptions scales (i.e., good teaching, clear goals and standards, appropriate assessment, appropriate workload, and emphasis on the independence) and approaches to learning.

Despite the usefulness of the CEQ in assessing students' perceptions of the learning environment under the relational student learning perspective, the development and use of the CEQ is largely restricted to the traditional face-to-face teaching and learning contexts. This limitation makes it unsuitable to be used in the modern higher educational contexts, in which the adoption of Internet and Web-based technologies has become an integral part of the students' learning experience. Through widespread use of teaching and learning resources beyond printed materials, such as multimedia, internet, mobile applications, digital games, a large number of courses offered in tertiary institutions are designed as blended learning, which includes a significant proportion of online mode beyond sitting in traditional face-to-face lectures and tutorials (Castle & McGuire, 2010; Singh, 2003; Verkroset, Meijerink, Lintsen, & Veen, 2008).

In blended learning, where students move back and forth across in-class and on-line contexts, their perceptions of the extent of the integration between face-to-face and online learning has been found to shape the way students approach learning and use online learning technologies (e.g., Ellis & Bliuc, 2016) and the level of engagement with a variety of online learning activities and in-class activities using online materials (e.g., Ellis & Han, 2018). Notwithstanding the wide spread of blended learning in higher education, there is yet to be reliable and valid instruments, which measure students' perceptions of the blended learning environment adopting the relational student learning perspective. Hence, the aim of our study

was to provide an initial attempt to construct and validate a Perceptions of the Blended Learning Environment Questionnaire (PBLEQ). The PBLEQ was designed to assess key aspects of the blended learning environment which shape university students' perceptions of the online part of a blended learning environment at course level. In the following sections, we first report the conceptual development of the PBLEQ, which are followed by the validation of the questionnaire through three studies.

### **The Conceptual Development of the PBLEQ**

Evaluating learning in blended environments is a complex goal, as students are required to move back and forth between face-to-face and online contexts when engaging in their learning activities. One of the main difficulties is the *relational* nature of the online part to the whole course. One learning activity can commence in class, continue online out of class, and then require students to finish it off in class. Evaluating this part/whole relationship is difficult. For this reason, the questionnaire investigates aspects that deal with the online part in the context of the *whole* course. Another of the difficulties is the range of issues that could legitimately be involved in how students perceive the online part of their course.

To address these difficulties and to construct and test items that might make up coherent and illuminative scales, interviews with students were undertaken, fully transcribed, and analyzed. The key questions of the interview were about the benefits and challenges the students perceived to be the most important for their learning in blended environments. Through thematic analyses of the transcripts, the key benefits and challenges of the online part of blended courses raised by the student interviews were (1) the relevance of the online activities/resources to face-to-face learning, (2) student online contributions, and (3) online workload.

Using the three broad categories derived from the interviews, we compiled the items pool by consulting the literature and questionnaires which adopted a relational perspective. Items for the integration and contribution categories were informed by relational student learning research

(e.g., Biggs, 2011; Prosser & Trigwell, 1999; Ramsden, 2003), without actual items being adapted from questionnaires. Some items from the category of the online workload were adapted from the workload scale in the CEQ (Ramsden, 1991). For example, the item “The workload for the online activities was too heavy” was adapted from “The workload was too heavy” in CEQ. The item pools consisted of 13 items assessing students’ view of the level of integration of the online component with the whole course; 7 items assessing students’ perceived value of online contributions by other students in the course; and 6 items assessing students’ perceptions of course workload in relation to the online activities.

### **The Validation of the PBLEQ**

The validation of the PBLEQ was undertaken in three studies. Study 1 explored the factor structure of the PBLEQ, which included the item analysis, EFA, and scale reliability analysis. The EFA and reliability analysis allowed students’ perceptions of the aspects of blended learning environment to be represented by a set of observed scale scores, thus they were able to show evidence of validity through scoring inferences to a certain extent (Kane, 2013). Study 2 evaluated the factor structure of the PBLEQ retained from the EFA against alternative models using CFAs among two separate cohorts of students studying in courses from either humanities/social sciences disciplines or from sciences/engineering disciplines. Study 3 tested the factorial invariance of the PBLEQ by performing a series of invariance tests across the above two cohorts of students. The details of the samples, methods of analysis, and the results of each study are described in the following.

### **Study 1 – Exploring Factor Structure of the PBLEQ**

#### **Sample of Study 1**

The sample of Study 1 was 397 first year students at a research-intensive Australian university. They were enrolled in a compulsory course – Human Biology. Among them, 272

were females, and 124 were males, with 1 missing gender information. Their ages ranged from 18 to 53 ( $M = 19.69$ ;  $SD = 3.47$ ).

### **Analysis of Study 1**

**The item analysis.** The item analysis was performed by calculating the means, standard deviations, skewness, and kurtosis of each item to check whether the items had ceiling and floor effects, and whether the items had a reasonable dispersion.

**The EFA.** Following the item analysis, an EFA was conducted using principal axis factoring method to explore the structure of the questionnaire. An oblique rotation was adopted following Tabachnick and Fidell's (2007) suggestion that "Perhaps the best way to decide between orthogonal and oblique rotation is to request oblique rotation [e.g., direct oblimin or promax from SPSS] with the desired number of factors [see Brown, 2009] and look at the correlations among factors..." (p. 646). To retain appropriate items, the items with factor loadings less than .30 within a factor, and cross loading items, which load at .32 or higher on two or more factors were deleted (Tabachnick & Fidell). To determine the number of factors, apart from considering the interpretability of the solution (Preacher & MacCallum, 2003), we performed the Velicer's minimum average partial analysis (MAP; Velicer, 1976), and parallel analysis (Horn, 1965), both of which were run in IBM SPSS 24.0 utilizing the map.sps and the rawpar.sps scripts, respectively (O'Connor, 2000).

**The scale reliability analysis.** The final analysis of Study 1 was to conduct Cronbach's alpha scale reliability analyses to evaluate the internal consistency of the scores of the retained scales.

### **Results of Study 1**

**Results of item analysis.** The means of all the items ranged from 1.90 to 3.84 out of 5. The values of skewness were between -0.85 and 1.11, and kurtosis ranged between -0.71 and 0.61. All the skewness and kurtosis were within the acceptable values  $\pm 2$  (Gravetter & Wallnau,

2014; Trochim & Donnelly, 2006), indicating that none of the items had a substantial ceiling or floor effect. The standard deviations of all the items were close to 1 (between 0.91 and 1.14), suggesting that the items had satisfactory dispersion. Thus, all the items were retained for the EFA.

**Results of the EFA.** The Kaiser-Meyer-Olkin (KMO) was .86, Bartlett's test of sphericity's chi-square was significant:  $\chi^2(120) = 3109.67, p < .01$ , which verified the sampling adequacy for performing an EFA (Field, 2013; Hutcheson & Sofroniou, 1999). Using the criteria stated above, 10 items, which either had low loadings within a factor or cross loaded on two or more factors were removed.

For the MAP test, the map.sps script produced average squared partial correlation and average 4th power partial correlation using the original (Velicer, 1976) and the revised (Velicer, Eaton, & Fava, 2000) MAP tests respectively. The smallest average squared partial correlation was .03 and the smallest average 4th power partial correlation was .00, both of which were the third factor, indicating the number of components was three. We performed the parallel analysis adopting principal axis/common factor analysis approach based on permutations of 1000 datasets. The analysis generated eigenvalues from the raw data along with the mean eigenvalues and eigenvalues representing the 95<sup>th</sup> percentile of the permutations. We compared the eigenvalues of the raw data and those of the 95<sup>th</sup> percentile estimates (eigenvalues raw vs. 95<sup>th</sup> percentile: factor 1: 4.57 vs. 0.50, factor 2: 2.90 vs. 0.40, factor 3: 1.69 vs. 0.33, factor 4: raw 0.18 vs. 0.27). The comparison also suggested a three-factor solution because the eigenvalues of the raw data was below the 95<sup>th</sup> percentile estimates in the fourth factor.

The results of the EFA showed that the eigenvalues of the three scales were 3.15, 3.35, and 2.79 respectively, and they could account for approximately 58% of the variance. The mean values of the 16 items ranged from 1.90 to 3.70, and the standard deviations were from 0.91 to



1.14. The factor loadings of the pattern matrix, means, and standard deviations of each item were presented in Table 1.

[insert Table 1 about here]

From Table 1, we can see that *the perceptions of integration between face-to-face and online learning* scale consists of 6 items, which evaluate students' perceptions of the level of the integration between face-to-face learning component and the online learning component, with factor loadings ranging between .67 and .76. *The perceptions of the online contributions* scale, which assesses students' perceptions of the values associated with online contributions from their classmates, has 5 items with factor loadings from .71 to .82. *The perceptions of the online workload* scale, which examines students' perceptions on the level of online workload in the blended course, encompasses 5 items with factor loadings from .59 to .89.

The correlation analyses showed that the perceptions of integration between face-to-face and online learning scale was positively related to the perceptions of the online contributions scale ( $r = .44, p < .01$ ), and to the perceptions of the online workload scale ( $r = .16, p < .01$ ), whereas the perceptions of the online contributions scale was negatively associated with the perceptions of the online workload scale ( $r = -.10, p < .05$ ). The results of the correlations further supported the adoption of the oblique rotation in the EFA. We also calculated a total perception score and examined the correlations between the total score and the scores of the three scales, which were .77 (the perceptions of integration between face-to-face and online learning), .66 (the perceptions of the online contributions), and .45 (the perceptions of the online workload), respectively. The strength of the correlations provided some evidence to scoring inferences of both the coherence and independence of the scores.

**Results of the scale reliability analysis.** The Cronbach's alpha coefficients in Table 1 show that all the three scales were highly reliable: the perceptions of integration between face-to-face and online learning scale: .88, the perceptions of the online contributions scale: .89, and

the perceptions of the online workload scale: .84. These values were well above the recommended acceptable value of .70 (Field, 2013). These results suggested that the scores maintained good internal consistency and were able to be generalizable, providing additional evidence to scoring inferences.

## **Study 2 – Evaluating the Factor Structure of the PBLEQ against Alternative Models**

### **Sample of Study 2**

The sample of Study 2 was two cohorts of students recruited from the same Australian university as the sample of Study 1. Cohort 1 had 609 students, who studied a course in humanities/social sciences, and they were between 18 to 51 years old ( $M = 21.17$ ;  $SD = 3.09$ ). Cohort 2 had 592 students, who were enrolled in a course in sciences/engineering, and their ages were between 17 to 61 ( $M = 21.76$ ;  $SD = 5.09$ ).

The theoretical basis for the two broad grouping of students was from “Becher-Biglan typology” in which humanities/social sciences are categorized into soft academic disciplines, whereas sciences/engineering are categorized into hard academic disciplines (Becher, 1989, 1994; Becher & Trowler, 2001; Biglan, 1973a, b; Neuman, Parry, & Becher, 2002).

### **Analysis of Study 2**

We conducted two sets of CFAs of three models separately for the two cohorts of students. For each set, the first CFA tested the correlated three-factor model as based on the results of the EFA in Study 1. The second CFA tested a second-order model, in which the three factors were integrated into a higher-level factor of the perceptions. The third CFA was a bi-factor model (Reise, Moore, & Haviland, 2010), which specified both a general factor of the perceptions that was hypothesized to account for the commonality of all the items; and the three specific perceptions factors with regard to the face-to-face and online integration, the online contributions, and the online workload of the perceptions, each of them was hypothesized to account for the unique influence over the general factor of the perceptions.

The CFAs were conducted using Mplus version 7.0. To evaluate the CFA models, we considered three fit statistics as primary indicators of model fit – the Tucker-Lewis Index (TLI, Tucker & Lewis, 1973), the Comparative Fit Index (CFI, Bentler, 1990), and the root mean square error of approximation (RMSEA, Browne & Cudeck, 1993) due to the sensitivity of the chi-square statistics to sample size. According to Bentler, the values of CFI and TLI higher than .900 are generally considered as an acceptable fit to the data. Browne and Cudeck suggest that RMSEA values below .060 are indicative of a good fit. Support for the CFAs also requires that the scores of scales are reliable (i.e.,  $\alpha = .70$  or above; Jöreskog & Sörbom, 2005).

## Results of Study 2

[insert Table 2 about here]

*Results of CFAs for the humanities/social sciences cohort.* Table 2 shows the results of the model fit of the three CFAs for the humanities/social sciences students. Even though the correlated three-factor model and the second-order model produced good fit, the bi-factor model yielded the best fit:  $\chi^2(88) = 215.487$ , CFI = .969, TLI = .957, RMSEA = .049, among the three CFAs. The correlated three-factor model and the second-order model had identical fit statistics because the second-order model was just-identified:  $\chi^2(101) = 313.911$ , CFI = .948, TLI = .938, RMSEA = .059. The results supported a bi-factor model for the PBLEQ among the humanities/social sciences cohort. The values of the Cronbach's alpha of the general perception factor was .85, the perceptions of integration between face-to-face and online learning scale was .85, for the perceptions of the online contributions scale was .88, and .79 for the perceptions of the online workload scale.

[insert Table 3 about here]

*Results of CFAs for the sciences/engineering cohort.* The results of the model fit of the CFAs for the sciences/engineering cohort are displayed in Table 3. Similar to the humanities/social sciences cohort, while the correlated three-factor model and the second-order

model generated acceptable and identical fit indices:  $\chi^2(101) = 276.708$ , CFI = .956, TLI = .947, RMSEA = .054; the bi-factor model produced the best fit:  $\chi^2(88) = 228.824$ , CFI = .964, TLI = .951, RMSEA = .052. All the scales showed good reliability: the general perceptions scale was .77, the perceptions of integration between face-to-face and online learning scale was .85, the perceptions of the online contributions scale was .89, and the perceptions of the online workload scale was .72. The bi-factor model for the PBLEQ was replicated with the sciences/engineering cohort. Hence, the bi-factor model was tested for its measurement invariance in Study 3.

### **Study 3 – Testing Factorial Invariance across Students from the Two Broad Disciplines**

#### **Sample of Study 2**

In Study 3, we used the sample of Study 2 to conduct a series of factorial invariance tests. Altogether there were 1201 students, around half of them (51%) studied a humanities/social sciences course, and the other half (49%) was enrolled in a sciences/engineering course.

#### **Analysis of Study 3**

On the basis of the bi-factor model obtained in Study 2, we performed a series of measurement invariance tests to determine if the bi-factor model was equivalent across the two cohorts of students in the two broad academic disciplines. The invariance tests involve evaluating various levels of restricted models and proceed in a stepwise manner from less to more restrictive. Therefore, the invariance models are nested because the imposed constraints are progressively added (Brown, 2006; Byrne, 2016). We followed Brown's recommended procedure for performing invariance tests by starting from separate evaluation of the CFA model in each group (this was carried out in Study 2) followed by a configural CFA (Sabiston et al., 2010; Van de Schoot, Lugtig, & Hox, 2012). The configural model, which is the least restricted model, tests whether the factor structures are identical across groups. Following the configural model, we tested whether factor loadings were equal in the metric model. We then constrained intercepts to be equal, referred to as the scalar model. The last model constrained residual

variance to be equal. Although it is possible to constrain other parameters to be equal (e.g., factor covariance), Brown (2006) argues that these invariance tests are not essential because the differences may reflect the real differences across groups.

There are two ways to assess the fit of the nested models for the invariance tests: to compare the chi-square statistics or the goodness-of-fit indices between the model with additional constraints and the less restricted model (Byrne & Stewart, 2006; Cheung & Rensvold, 2002). However, due to sensitivity of  $\Delta\chi^2$  to the sample size, Cheung and Rensvold recommended the  $\Delta CFI$ . According to Cheung and Rensvold, when the  $\Delta CFI$  is less than or equal to .010 the specified equal constraints are tenable, and when the  $\Delta CFI$  is greater than .010 between two nested models, the more constrained model is rejected.

### Results of Study 3

[insert Table 4 about here]

The results of the invariance tests are presented in Table 4, which shows that the configural model resulted in a good fit:  $\chi^2(176) = 444.092$ ,  $CFI = .967$ ,  $TLI = .954$ ,  $RMSEA = .050$ , suggesting the equal factor structure across the two cohorts. Because the metric model also produced an appropriate fit:  $\chi^2(204) = 520.886$ ,  $CFI = .961$ ,  $TLI = .954$ ,  $RMSEA = .051$ , and the  $\Delta CFI$  was .006, which was less than the .010, the equal factor loadings across the two cohorts of students were supported. A good fit of the scalar model was also attained:  $\chi^2(216) = 586.596$ ,  $CFI = .954$ ,  $TLI = .949$ ,  $RMSEA = .053$ , with  $\Delta CFI$  being .007, supporting the intercepts invariance across the two cohorts. The results of the residual invariance model did not produce appropriate fit:  $\chi^2(232) = 1125.936$ ,  $CFI = .889$ ,  $TLI = .885$ ,  $RMSEA = .080$ , suggesting a possible variance of the residual parameter. However, Byrne (1994) and Marsh (1994) argued that fixing the residual parameter to be equal is an overly restrictive practice for evaluating invariance. Overall, the results of our measurement invariance tests demonstrated that the bi-

factor model of the PBLEQ is consistent across the two cohorts of students studying courses in the two broad academic disciplines.

### **Discussion**

The purpose of the current study was to describe the process of the initial development and investigation of the psychometric properties of the PBLEQ. The formal assessment of the psychometric properties contributes to our understanding of how the items in the questionnaire load together and provides some evidence for the reliability and validity of scores of this assessment tool. The relations between the full score of the general perceptions scale and the three specific subscale scores showed the utility of the full score and the distinctiveness of the subscales, which can be used to measure the general levels of students' perceptions, and the levels of perceptions of the three distinct aspects, of the online part of learning experience in the blended environment. It is worth noting that use of the PBLEQ is targeted at the level of a course rather than the level of a degree, because within a degree, not all courses are necessarily designed as a blended learning experience. According to the results, the key aspects for the online part of the learning environment in a university course rated by students were the extent of integration of the online resources into their learning activities, assessment, classwork; the usefulness of the online contributions from other students to their own learning, and whether or not the online workload was prohibitive in relation to total course workload. Its development and validation addressed the lack of an illuminative instrument which can reveal how positive and negative perceptions of these dimensions may be investigated for their interrelatedness with learning outcomes and other aspects of the learning experience.

In this study, we adopted the relational student learning perspective to construct the PBLEQ. While the EFA generated a three-correlated structure of the 16-item PBLEQ, the CFAs supported the bi-factor model over the correlated three-factor model and the second-order model consistently across the two cohorts of students studying a course either in humanities/social

sciences or sciences/engineering, suggesting the stability of the bi-factor model of the PBLEQ. The CFAs allowed to compare the alternative factorial structures of the PBLEQ in addition to the initial factorial structure obtained from the EFA, hence, they were able to provide more informative and rigorous examination of the factorial structure (Raykov, 2001). The CFAs consistently supported the bi-factor model over the correlated three-factor model and the second-order model. The final version of the 16-item PBLEQ is represented by a general factor of perceptions with three specific factors on the three aspects of online part of learning in the blended environment, namely (1) the perceptions of integration between face-to-face and online learning; (2) the perceptions of the online contributions; and (3) the perceptions of the online workload. The fit of the bi-factor model and the reliability of one general and the three specific factors in the model were consistently produced across the two groupings of students, demonstrating the stability of the structure and reliability of the PBLEQ.

The series of measurement invariance tests on examination of invariant factor structure, factor loadings, and intercepts across the two cohorts of students demonstrated that students shared the same interpretation of the items with regard to the learning environment in the blended courses of different academic disciplines. Because of the identical pattern of factor-indicator relationships, factor loadings, and intercepts, the factor scores from the two cohorts can be legitimately compared (Brown, 2006). The examination of the measurement invariance ensured the potential use of the PBLEQ in various academic disciplines in higher education.

### **Limitations and Directions for Future Research**

The current study has a number of limitations, which should be addressed in future research. First, we developed the dimensions to be covered for the aspects of online part of learning in the blended contexts from our interviews with students, who were all from the same institution. Thus, the three dimensions in the PBLEQ by no means represent a comprehensive coverage of all aspects of the online part in blended course designs. Other aspects, such as

perceptions of the physical learning spaces and perceptions of online interactions, and perceptions of the usability of online design, may be included in the further development and validation of the PBLEQ.

Second, the current study did not assess the validity of the PBLEQ scores based on relationships to the scores of other standardized or validated scales because of a lack of appropriate scales in the literature. This motivated us to develop the PBLEQ as an initial attempt to address the unavailability of the measurement tools for perceptions of blended learning environment from a lens of the relational student learning.

Another limitation lies in the sampling that all the samples were recruited from one single research-intensive Australian university, whose blended teaching and learning contexts and requirements may be different from other universities in Australia and in other countries. Future research can draw students from other populations to further validate the PBLEQ. While our study addressed the validity through scoring inferences, we did not examine the validity through generalization and extrapolation inferences as proposed in the test validation framework (e.g., Kane, 2013; Whitney, Cheng, Brodersen, & Hong, 2018). Future work should be carried out on further building the validity argument of the PBLEQ through generalization and extrapolation inferences.

Notwithstanding these limitations, our research has made a useful first attempt to address a gap in the relational student learning research in higher education by developing an appropriate instrument that measures students' perceptions of the online part of learning environment in blended contexts so that students' blended learning experience can be better understood.

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Table 1

*The Results of the EFA*

| factors                                                                       | item description                                                                                   | rotated factor loadings |             |             | <i>M</i> | <i>SD</i> |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------|-------------|-------------|----------|-----------|
| the perceptions of integration between face-to-face and online learning (.88) | The online activities helped me to understand the lectures in my course.                           | <b>0.73</b>             | 0.20        | -0.01       | 3.60     | 1.11      |
|                                                                               | The online activities seemed to be well integrated with the assessment.                            | <b>0.67</b>             | 0.14        | 0.10        | 3.01     | 1.04      |
|                                                                               | The resources on the course website helped me to understand ideas in class.                        | <b>0.75</b>             | 0.10        | 0.06        | 3.58     | 0.96      |
|                                                                               | Studying online materials helped with the assessment in the course.                                | <b>0.71</b>             | 0.15        | -0.03       | 3.50     | 1.05      |
|                                                                               | I found the resources on the course website very useful for my learning.                           | <b>0.73</b>             | 0.19        | 0.01        | 3.66     | 0.98      |
|                                                                               | The ideas online in this course related to the ideas in class.                                     | <b>0.76</b>             | 0.12        | 0.00        | 3.70     | 0.91      |
| the perceptions of the online contributions (.89)                             | Other students' online contributions helped me understand my ideas from a new perspective.         | 0.14                    | <b>0.71</b> | -0.13       | 3.10     | 1.11      |
|                                                                               | Online contributions from others prompted me to reflect more on the ideas in this course.          | 0.21                    | <b>0.80</b> | -0.06       | 3.14     | 1.12      |
|                                                                               | The online contributions from other students helped develop my understanding of particular topics. | 0.25                    | <b>0.82</b> | -0.08       | 3.05     | 1.10      |
|                                                                               | Online contributions by students in this course motivated me to think about things more.           | 0.14                    | <b>0.78</b> | -0.02       | 2.77     | 1.07      |
|                                                                               | Online contributions by students in this course prompted me to engage more.                        | 0.14                    | <b>0.73</b> | -0.07       | 2.77     | 1.04      |
| the perceptions of the online workload (.84)                                  | *I needed more time to do my online activities.                                                    | -0.02                   | -0.16       | <b>0.63</b> | 1.90     | 1.05      |
|                                                                               | *By the time I finished the online activities I was running behind.                                | 0.01                    | -0.05       | <b>0.69</b> | 2.19     | 1.14      |
|                                                                               | *The online learning in this course took too much time.                                            | 0.12                    | 0.00        | <b>0.85</b> | 2.17     | 1.10      |
|                                                                               | *The workload for the online activities was too heavy.                                             | 0.12                    | 0.06        | <b>0.89</b> | 2.18     | 1.11      |
|                                                                               | *A better balance between the online activities and the other tasks would help my workload.        | -0.07                   | -0.14       | <b>0.59</b> | 1.97     | 1.02      |

Note: \*negatively-worded items were reversed.



Table 2

*Goodness of Fit of CFAs for Humanities/Social Sciences Students*

| model                         | $\chi^2$ | df  | CFI  | TLI  | RMSEA |
|-------------------------------|----------|-----|------|------|-------|
| correlated three-factor model | 313.911  | 101 | .948 | .938 | .059  |
| second-order model            | 313.911  | 101 | .948 | .938 | .059  |
| bi-factor model               | 215.487  | 88  | .969 | .957 | .049  |

*Note:*  $N = 609$ ; CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = Root mean square error of approximation.

Table 3

*Goodness of Fit of CFAs for Sciences/Engineering Students*

| model                         | $\chi^2$ | df  | CFI  | TLI  | RMSEA |
|-------------------------------|----------|-----|------|------|-------|
| correlated three-factor model | 276.708  | 101 | .956 | .947 | .054  |
| second-order model            | 276.708  | 101 | .956 | .947 | .054  |
| bi-factor model               | 228.824  | 88  | .964 | .951 | .052  |

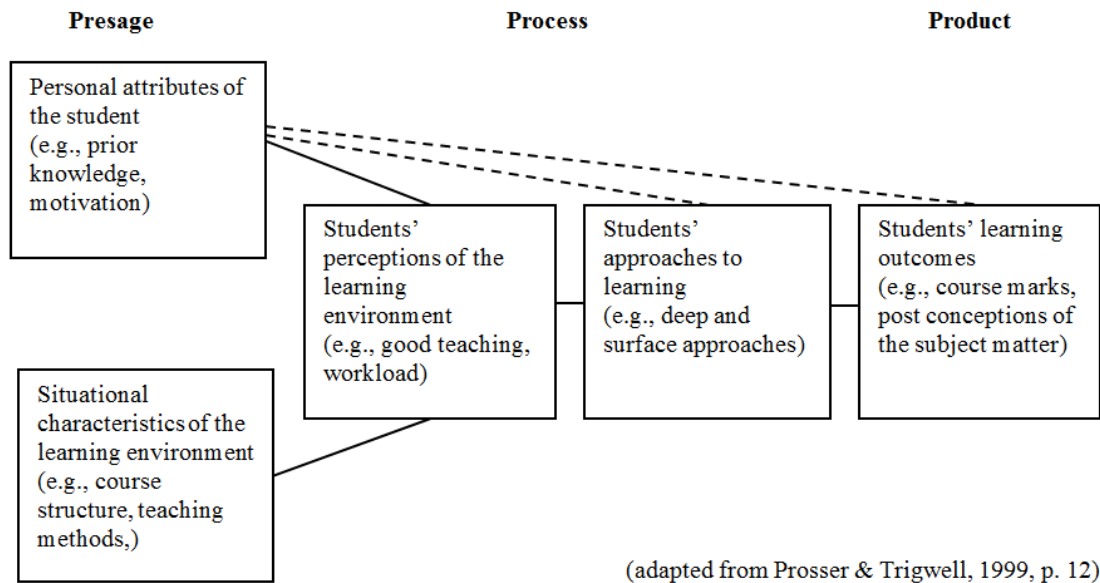
*Note:*  $N = 592$ ; CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = Root mean square error of approximation.

Table 4

*Goodness of Fit of Invariance Tests*

| model               | $\chi^2$ | df  | CFI  | TLI  | RMSEA | $\Delta$ CFI |
|---------------------|----------|-----|------|------|-------|--------------|
| configural          | 444.092  | 176 | .967 | .954 | .050  | ---          |
| metric              | 520.886  | 204 | .961 | .954 | .051  | .006         |
| scalar              | 586.596  | 216 | .954 | .949 | .053  | .007         |
| residual invariance | 1125.936 | 232 | .889 | .885 | .080  | .065         |

*Note:*  $N = 1201$ ; CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = Root mean square error of approximation.



*Figure 1. Presage-Process-Product Model of student learning*